

David Zech

Software Security Engineer

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EXPERIENCE

Apple, Inc.

June 2020 – Present

Senior Software Engineer, Security

Cupertino, CA

- Co-inventor and architect for the Private Cloud Compute (PCC) system for Apple Intelligence. Designed and implemented remote attestation and verifiable software transparency solutions to securely and privately offload Apple Intelligence requests to Apple's data centers. Authored several sections of the Private Cloud Compute Security Research Guide that is publicly available for all users.
- Project lead of the CloudAttestation framework, the key component that delivers the privacy and security guarantees of all PCC-offloaded Apple Intelligence prompts by ensuring customer prompts are encrypted so only genuine Apple Silicon Servers can decrypt them. Written in Swift and shipped in all major Apple OS platforms.
- Developed the Apple Transparency Log auditing features for the Private Cloud Compute Virtual Research Environment which shipped in all macOS 15+ installations, allowing security researchers to verify the state of the Transparency Log using a local copy of the append-only Merkle Tree.
- Core developer of FileVault, the full disk encryption technology for macOS. Added the ability to unlock FileVault enabled disks over the network with OpenSSH, and led the migration of millions of Mac devices off the legacy iCloud-based recovery-key escrow to an end-to-end encrypted recovery system.
- Developed several new cryptographic features for the AppleKeyStore, the primary firmware component that underpins the hardware-backed data protection features for all shipped Apple products, including Device Activation, Attestation, Passcode Protection, Biometric Authentication, and per-file encryption.
- Project lead for TPM Authority, a key residency attestation service for Trusted Platform Modules certified for use within Apple. Bootstraps hardware-backed cryptographic identity for all of Apple's x86 servers.

Uber Technologies, Inc.

November 2019 – June 2020

Software Engineer II

Palo Alto, CA

- Built the uProtect Licensing Services, a collection of in-house public key infrastructure services to provide external clients and devices short-term network access credentials. Enabled Uber ATG's fleet of self-driving cars to connect and authenticate to Uber's edge network securely over mutual TLS.

Sylabs, Inc.

March 2019 – October 2019

Software Engineer

Remote

- Developed features for the Singularity Container Engine, a Docker compatible container runtime, and supported the development of the enterprise on-premise version of Sylabs Cloud.

Intel Corporation

June 2015 – February 2019

Software Development Engineer

Folsom, CA

- Worked on applications leveraging Trusted Execution Technology and Trusted Platform Modules, and led the teamwide switch from Java to Go as the primary coding language.

SKILLS

Languages: Swift, Go, Java, Python, C/C++, C#, Objective-C, Javascript

Technologies: Kubernetes, gRPC, Protobuf, Temporal, Grafana, Telegraf, Prometheus, iOS/macOS SDK, .NET MongoDB, SQL, HTML5, CSS, Node.js

EDUCATION

University of California, San Diego

2018

B.S. Computer Science

GPA: 3.8